



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO UMA FUEL AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

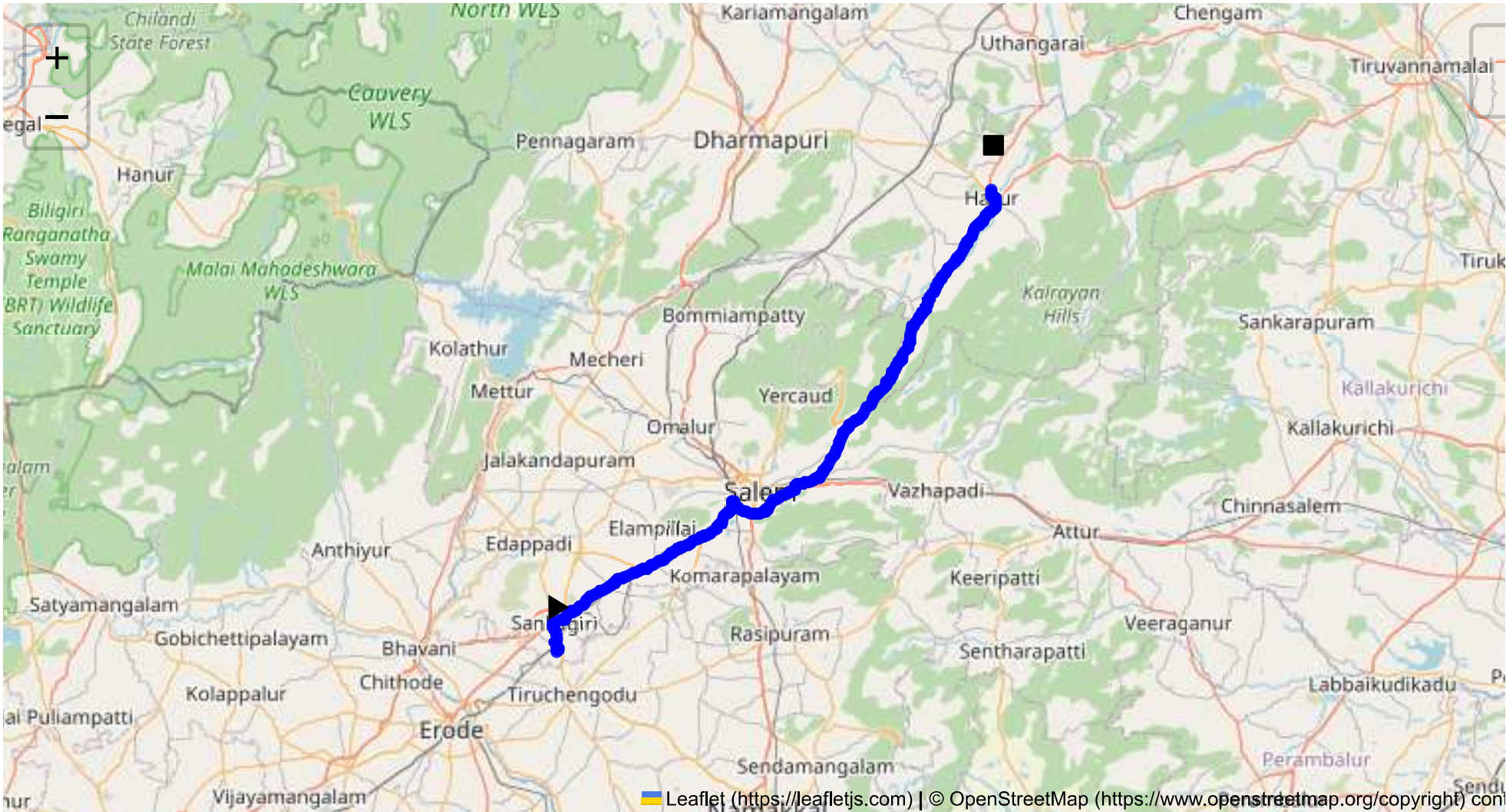
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 107.93 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours
Start: (11.4381, 77.8734)
End: (12.066818, 78.47683)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri, to 1D, Mobirippatti, Harur spans approximately 107.93 kilometers. Starting from Sangagiri, the route primarily follows NH44, then diverts onto the Salem-Tirupattur-Vaniyambadi Rd (SH18). This passage traverses through urban, semi-urban, and rural areas, requiring navigating varied types of roadways.

2. Weather Conditions and Potential Hazards

Tamil Nadu generally experiences hot and dry conditions with monsoon rains between October and December. Potential weather hazards include heavy rainfall leading to slippery roads, low visibility, and occasional flooding, especially during the monsoon.

3. Traffic Patterns

The route experiences peak traffic during morning hours (7 AM - 10 AM) and evening (6 PM - 9 PM) due to commuter movement. Salem city is a notable congestion point, and traffic tends to slow around major junctions like the Bangalore Bypass Road intersection on NH44.

4. Road Quality and Infrastructure

The NH44 is generally well-maintained with smooth surfaces, whereas SH18 may experience variable road conditions, including patches with potholes and narrow stretches. Road infrastructure is moderate, with an adequate number of signage and signals.

5. Alternative Routes for Emergencies

Alternatives include taking NH544 from Sankari to Salem and then SH82 towards Harur. Another is using Salem's Vellore Road (NH179A) to connect to Harur. These alternatives should be considered only if major obstructions are encountered on the primary route.

6. Local Regulations on Hazardous Material Transport

Transport of hazardous materials requires permits. Vehicles should adhere to speed limits, not exceed cargo limits, and follow designated routes. Nighttime restrictions might be in place in urban areas to minimize risks.

7. Historical Incidents

There have been incidents of vehicle overturns and collisions, mainly attributed to driver error and mechanical failure. Zones with dense traffic and sharp bends on SH18 are especially prone to such incidents.

8. Environmental Considerations

The route passes close to Koundampatti Reserve Forest and other ecologically sensitive areas. Care should be exercised to prevent pollution and disturbances, especially in these zones.

9. Communication Coverage

While most of the route maintains good cellular coverage, some stretches, particularly rural patches on SH18, may have intermittent signals. Critical for continuous communication with dispatchers or emergency services.

10. Estimated Emergency Response Times

Urban areas like Salem may see response times of 15-30 minutes, while rural segments could encounter delays up to an hour or more due to distance and road conditions.

11. Summary of Risk Assessment

In summary, the route involves moderate risk, primarily affected by variable traffic volumes, road conditions, and environmental sensitivities. Major risks include monsoon-related weather hazards and potential traffic delays in metropolitan zones. Proper compliance with safety regulations and preparedness for emergency rerouting can mitigate these risks. Vehicles should be equipped with GPS, communication devices, and have contingency plans in place.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45000, 77.87213	30 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.45309, 77.87325	30 KM/Hr
7	Turn	Medium	11.46400, 77.87323	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.53501, 77.95707	30 KM/Hr
13	Turn	Medium	11.53601, 77.95856	30 KM/Hr
14	Turn	High	11.64172, 78.11817	15 KM/Hr
15	Turn	Medium	11.64216, 78.11875	30 KM/Hr
16	Turn	High	11.64195, 78.12129	15 KM/Hr
17	Turn	High	11.64120, 78.12168	15 KM/Hr
18	Turn	Medium	11.63948, 78.12081	30 KM/Hr
19	Turn	Medium	11.66366, 78.20649	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	Medium	11.67129, 78.22978	30 KM/Hr
21	Turn	High	11.67120, 78.23136	15 KM/Hr
22	Turn	Medium	11.67198, 78.23148	30 KM/Hr
23	Turn	Medium	11.67440, 78.23914	30 KM/Hr
24	Turn	Medium	11.80440, 78.33325	30 KM/Hr
25	Turn	Medium	11.81235, 78.34084	30 KM/Hr
26	Turn	Medium	11.83482, 78.35294	30 KM/Hr
27	Turn	Medium	11.84738, 78.36062	30 KM/Hr
28	Turn	Medium	11.99554, 78.44553	30 KM/Hr
29	Turn	Medium	12.01577, 78.45673	30 KM/Hr
30	Turn	Medium	12.04808, 78.48399	30 KM/Hr
31	Turn	Medium	12.04979, 78.48303	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
21	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
22	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
23	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
24	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
25	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
26	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
27	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
28	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
29	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
30	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
32	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
34	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
35	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
43	hospital	Nanjaya Hospital	12.0550888, 78.4821929	30 km/h	Medium
44	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
45	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
20	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
31	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
33	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
36	school	Indian Matriculation Higher secondary school	12.006299, 78.4531204	30 km/h	Medium
37	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
38	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
39	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
40	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
41	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
42	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64216, 78.11875



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64195, 78.12129



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63948, 78.12081



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66366, 78.20649



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67129, 78.22978



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67120, 78.23136



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67198, 78.23148



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67440, 78.23914



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80440, 78.33325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81235, 78.34084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83482, 78.35294



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84738, 78.36062



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99554, 78.44553



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.01577, 78.45673



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04808, 78.48399



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04979, 78.48303